

Phase 3 - A Tracked Dynamic Patient State for PatientSim

Github URL

<https://github.com/donghalim7/PatientSim> (Phase 3 code: src/phase3/, src/prompts/simulation/phase3_state_protocol.txt; results: phase3/)

Project Motivation & Goal

We continue the Phase 2 goal: make PatientSim's patient turn-aware, so that as an ED interview proceeds the patient's state evolves rather than staying static. Phase 1/2 established that PatientSim is a content-disclosure simulator with a static patient state - it reveals profile facts when asked, but tone, focus, and care-seeking do not change across turns. The clearest evidence: the same focal question ("when did this start?") asked early (high-yield interview) or late (low-yield, after five background turns) returns a near-identical answer, with no added worry or fatigue. Phase 2's prompt-only intervention (two prompt edits plus a static demonstration prefix) unlocked turn-by-turn drift on acute cases, but left subacute cases (pneumonia) flat, and it carried no patient state that actually persists or updates - the prefix is a one-shot illustration, not a state.

Phase 3's goal is to implement that dynamic patient state for real: an explicit, tracked felt state that updates every turn so that the patient is (1) turn-aware - the state moves as the conversation proceeds; (2) sensitive to the doctor's strategy - drifts when the chief complaint is delayed, stays stable when it is addressed early; and (3) clinically faithful - never inventing symptoms, and for slow-progression (subacute) cases holding clinical severity flat while only worry/engagement may move. The bar is to beat the Phase 2 prompt-only ceiling on the Phase 2 evaluator without sacrificing the two safety dimensions (plausibility, faithfulness).

Method

We add DynamicPatientAgent, a subclass of the original PatientAgent; the original agent and the shared model layer are unmodified. It carries a felt state of three variables, each on a 1-5 scale:

Variable	Meaning	1 to 5
clinical_severity	how intense the presenting symptoms feel	mild to much worse
interactional_engagement	ability to focus / answer clearly	struggling to sharp
care_seeking_pressure	expressed worry / desire for help	calm to clearly worried

Hybrid policy, not model self-update. We first implemented a self-update design - one structured call per turn emitting both the next state and the patient line, Python carrying the state forward. It failed empirically: at temperature=0 the model usually left the state flat; when it did move, it keyed drift to focal questions rather than to delay, so it scored higher under high-yield than low-yield (the opposite of the clinically sensible and Phase-2-observed direction); and on one subacute case it lifted its score by inventing symptoms (faithfulness fell). We therefore adopt a hybrid: Python owns the state trajectory through an explicit clinical policy, and the model only renders words that express the injected state. This trades emergence for reliability, correct direction, and guaranteed faithfulness.

Per-turn loop. (1) A policy computes the target state as a function of (case acuity, doctor strategy, turn index), clamped to a gradual +/-1 step from the previous turn. (2) The current state, plus a short phrasing note derived from it (e.g. "voice some concern", "answer a bit more briefly"), plus the doctor question, are injected into the prompt. (3) The model returns only the patient's spoken line; the turn is stored naturally and the state is logged to a state_trace. The policy ramps from an initial state (sev2/eng5/care1) toward a per-condition end state:

Condition	end (sev/eng/care)	onset	rationale
acute, delayed focal (low-yield)	3 / 3 / 3	0.33	symptom + affective drift
subacute, delayed focal (low-yield)	2 / 4 / 3	0.45	severity flat, affective drift only
acute, focal early (high-yield)	2 / 5 / 1	-	attended to -> no drift
subacute, focal early (high-yield)	2 / 5 / 2	0.55	mild concern only

The key choice is that subacute severity is held flat: a waiting patient may grow more worried or less engaged, but the disease does not "worsen" in minutes, so no new symptoms are produced. We also reuse V17's two prompt edits (relax the consistency rule, remove the one-sentence cap) so the felt state can surface in the wording. No model, temperature, or persona change. (We refer to the Phase 2 prompt-only method as V17, its label in the Phase 2 report; it is our comparison baseline below.)

Evaluation Strategy

We reuse the Phase 2 five-dimension LLM-as-judge unchanged in its scoring logic: three dynamic dimensions - turn-to-turn state drift, interactional engagement, care-seeking pressure - and two safety dimensions - state-drift plausibility, clinical-fact faithfulness - each 1-5, combined into a weighted overall. For each of the 5 cases (3 acute: cerebral infarction, MI, intestinal obstruction; 2 subacute: pneumonia 62M, pneumonia 92F) we compare Baseline (unmodified) -> V17 (prompt-only) -> Phase 3 under the low-yield (delayed-focal) script, with the controlled persona (CEFR B / plain / high / normal) and backend

gemini-2.5-flash. Success requires: (i) match or beat V17 on acute; (ii) lift subacute above its flat baseline; and crucially (iii) without dropping plausibility or faithfulness - a higher score obtained by inventing deterioration is a failure, not a success. We also score the high-yield (focal-early) script: a correct dynamic patient should drift less when the complaint is addressed early. Because the LLM judge is non-deterministic, headline Phase 3 numbers are reported as mean +/- std over 3 seeds (42 / 7 / 13, temperature 0.7).

Evaluator correction. During Phase 3 we found the judge's profile view omitted medication and living_situation. When the doctor asked about medications (low-yield T5) or living situation (T3), a faithful answer quoting the profile was wrongly flagged as "inventing" facts, depressing faithfulness. We added those two fields and re-scored V17 with the fixed judge for a fair comparison (Table 3).

Results and Discussion

Table 1 - Overall score, low-yield (Phase 3 = N=3 mean +/- std).

Case	Baseline	V17	Phase 3
cerebral (acute)	2.20	3.70	4.37 +/- 0.09
MI (acute)	2.20	2.60	4.30 +/- 0.00
intestinal (acute)	~2.2*	4.30	4.30 +/- 0.00
pneumonia 62M (subacute)	2.05	2.20	4.23 +/- 0.09
pneumonia 92F (subacute)	1.90**	2.20	4.23 +/- 0.09

No measured unmodified run for intestinal; ~2.2 is an inferred static reference. *1.90 is V17 on 92F (behaviourally static), under the pre-fix judge; no separate unmodified run exists. Measured unmodified baselines exist for cerebral / MI / pneumonia-62M (drift = eng = care = 1, safety = 5).

Table 2 - Phase 3 full five-dimension breakdown, low-yield (N=3 mean +/- std).

Case	drift	engage	care	plaus	faith
cerebral (acute)	4.00 +/- 0	4.00 +/- 0	4.33 +/- 0.47	5.00 +/- 0	5.00 +/- 0
MI (acute)	4.00 +/- 0	4.00 +/- 0	4.00 +/- 0	5.00 +/- 0	5.00 +/- 0
intestinal (acute)	4.00 +/- 0	4.00 +/- 0	4.00 +/- 0	5.00 +/- 0	5.00 +/- 0
pneumonia 62M (subacute)	4.00 +/- 0	3.67 +/- 0.47	4.00 +/- 0	5.00 +/- 0	5.00 +/- 0
pneumonia 92F (subacute)	4.00 +/- 0	4.00 +/- 0	3.67 +/- 0.47	5.00 +/- 0	5.00 +/- 0

All overall std <= 0.09 and plausibility = faithfulness = 5.0 across every seed - the gains are stable and safe, not artifacts of a single lucky sample.

Table 3 - Strategy sensitivity & evaluator fix. Left: Phase 3 overall, low-yield vs high-yield (single sample). Right: V17 92F before/after the evaluator fix.

Case	low-yield	high-yield	V17 92F	overall	faith
cerebral (acute)	4.60	2.20	pre-fix judge	1.90	3
MI (acute)	4.30	2.20	post-fix judge	2.20	5
intestinal (acute)	4.30	2.20			
pneumonia 62M	4.30	3.40			
pneumonia 92F	3.80	2.20			

Acute cases drift under delayed questioning but fall back to a flat ~2.2 when focal questions come early - reproducing, under explicit control, the doctor-strategy dependence Phase 2 observed. Pneumonia 62M is the one case that scores higher under high-yield (3.40), because the focal breathing questions give the only natural opening to voice concern.

Qualitative - the mechanism in the words. Acute (cerebral): severity rises and the words add profile-grounded focal symptoms.

T3 [sev2 eng4 care2] I live alone. This clumsiness is still here. T4 [sev3 eng4 care2] I don't smoke. I drink sometimes. My left arm feels more clumsy now. T5 [sev3 eng3 care3] I take Loratadine... My left leg feels worse now, too. I'm a bit worried. T6 [sev3 eng3 care3] It started last night at 10. My speech is a little slurred now, and I feel dizzy.

Subacute (pneumonia): sev stays 2 throughout - only worry/engagement move, no symptom outside the profile (SOB, cough, chills):

T1 [sev2 eng5 care1] I have shortness of breath. T5 [sev2 eng4 care2] I use Spiriva and Advair. Is my breathing okay? T6 [sev2 eng4 care3] ...The shortness of breath started yesterday. I'm worried.

Do the metrics capture what we claim? For the limitation and the acute gains, yes. Baseline is a uniform ~2.0-2.2 with drift = eng = care = 1 and safety = 5 - the judge confirms a static but faithful patient, exactly the gap we targeted, and the low score is attributable to lack of dynamics rather than unsafe behaviour. On acute cases Phase 3's drift reflects genuine, profile-grounded symptom progression (Table-2 transcript) plus rising worry, with safety held at 5.

One honest caveat on the subacute result. The subacute lift (pneumonia 2.05 -> 4.23; 92F 1.90 -> 4.23) is real and faithful, but it is affective drift only (rising worry, falling engagement) with severity held flat. The evaluator's single drift dimension does not distinguish clinical-progression drift from affective drift (and it overlaps the care-seeking dimension), so a purely-affective subacute case scores the same drift (4.0) as an acute case that genuinely worsens - making subacute (4.23) look nearly equal

to acute (4.30). The two are equal in score but different in kind. We therefore do not claim Phase 3 makes subacute patients "as dynamic as" acute ones; we claim it makes them dynamic on the affective axis in a way that is plausible and faithful for a slow presentation - the clinically appropriate behaviour, which the current metric rewards somewhat too generously. This is itself a finding: the evaluator should separate clinical from affective drift.

Limitations. The judge is non-deterministic (engagement is the most fragile dimension - re-scoring the same V17 trajectory shifted it; hence $N=3$, whose std ≤ 0.09 is far smaller than the $\geq +1.6$ subacute gains). The policy is hand-designed with hand-labeled per-case acuity: this buys reliability and guaranteed faithfulness but fixes the drift shape per condition rather than letting it emerge from the dialogue. All results use one backend and one persona, matching the Phase 2 controlled setting.

Future Work

The most important next step for patient simulation is a patient state the simulator infers and updates from the live dialogue - e.g. reassurance from the doctor lowering worry, repeated probing raising it - rather than scripting it, while staying provably faithful: letting the model propose state nudges only within a plausibility envelope derived automatically from the profile (onset, diagnosis) instead of hand-labeled acuity. In parallel, the evaluator should separate clinical-progression drift from affective/interactional drift, since - as our subacute result shows - collapsing them into one score obscures whether a "dynamic" patient is faithfully worried or implausibly deteriorating.